

1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like [instagram.com](https://www.instagram.com)). Stop the capture and note down the protocol names you see in the packet list.

I installed Wireshark and started capturing packets from my main network interface. Then, I opened a website such as Instagram and stopped the capture.

**Protocols I observed in the packet list:**

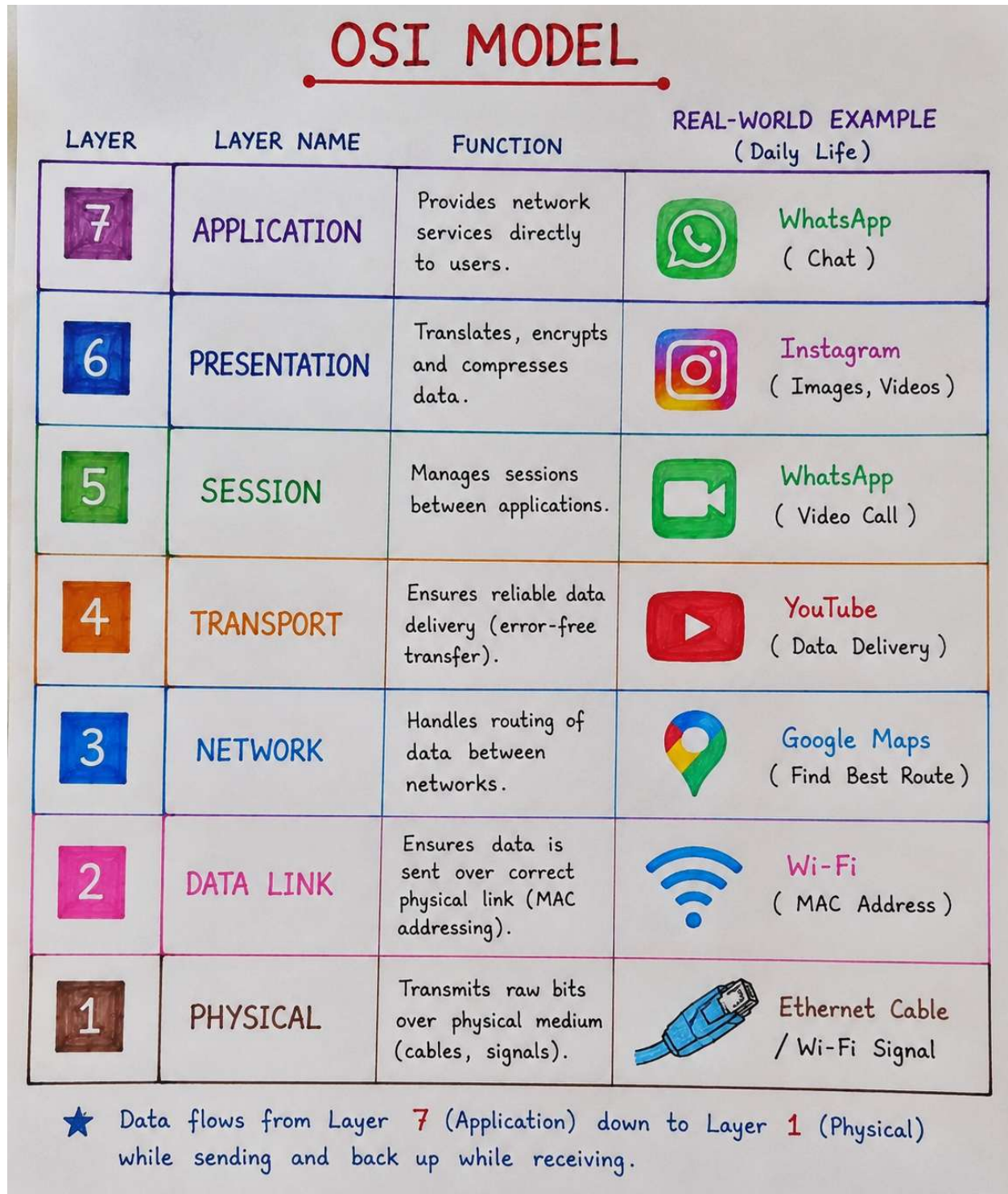
1. **DNS** – Used to find the website's IP address.
2. **TCP** – Helps establish a reliable connection.
3. **TLS** – Encrypts secure website communication.
4. **QUIC** – Used for fast and secure internet communication.
5. **UDP** – Transports data quickly between devices.

1. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.).  
  
**Hint:** Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

I selected 3 packets from my Wireshark capture:

| <b>Packet Protocol</b> | <b>OSI Layer</b>                   | <b>Simple Explanation</b>                        |
|------------------------|------------------------------------|--------------------------------------------------|
| <b>DNS</b>             | <b>Application Layer (Layer 7)</b> | Used to find the IP address of a website.        |
| <b>TCP</b>             | <b>Transport Layer (Layer 4)</b>   | Provides reliable data delivery between devices. |
| <b>IP</b>              | <b>Network Layer (Layer 3)</b>     | Helps send packets from one network to another.  |

2. Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).



4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words.<br><br><em><strong>Hint:</strong> Think about how your message is typed, packaged, sent over the internet, and received.</em>

When I send a message on **WhatsApp**, it travels through these OSI layers:

1. **Application Layer (Layer 7):** I type a message like "**Hello**" and press Send on WhatsApp.
2. **Transport Layer (Layer 4):** The message is divided into smaller parts and prepared for reliable delivery.
3. **Network Layer (Layer 3):** The data gets an **IP address** and routers find the path to the receiver's device.
4. **Data Link Layer (Layer 2):** The data is prepared to travel through the local network, such as **Wi-Fi**, using MAC addresses.
5. **Physical Layer (Layer 1):** The data is sent as **Wi-Fi radio signals or electrical signals** through cables.